ASSIGNMENT – MODULE 1

19-205-0802:MATERIALS MANAGEMENT (10.02.2026)

QUESTION 1

Case Study: Hospital Stores Department

A 300-bed multispecialty hospital faces frequent shortages of critical consumables and high holding costs due to poor materials planning.
The hospital administration wants to improve its materials management system.

Given Data:

The annual consumption and unit cost of selected medical items are:

Item	Annual Demand (units)	Unit Cost (₹)	Criticality
Syringes	18,000	12	Essential
Gloves	25,000	8	Vital
Antibiotic A	6,000	120	Vital
Cotton rolls	15,000	6	Desirable
Surgical sutures	4,000	250	Essential

Given:

- Ordering cost = ₹1,000 per order
- Inventory carrying cost = 20% of unit cost per year

(a) Explain the scope and objectives of materials management in a hospital setup.
(3 marks)

(b) Describe the purchase procedure followed by the hospital from indent to purchase order.
(4 marks)

(c) Perform ABC analysis for the given items.
Classify items into A, B, and C categories with justification.
(4 marks)

(d) Perform VED analysis based on criticality and identify AV, AE, BV items that need strict control.
(2 marks)

(e) For Antibiotic A, calculate Economic Order Quantity (EOQ) and comment on its significance.
(2 marks)

QUESTION 2

Case Study: Seasonal Retail Store

A retail store sells decorative lamps during the festival season.
The demand for the lamps is uncertain and follows the probability distribution below:

Demand (units)	100	150	200
Probability	0.3	0.4	0.3

Cost details:

- Purchase cost per lamp = ₹400
- Selling price per lamp = ₹650
- Unsold lamps can be sold later at ₹250 each

(a) Explain why this situation represents a static inventory problem under risk.
(3 marks)

(b) Identify and calculate:

- Excess cost per unit
- Shortage cost per unit
  (3 marks)

(c) Construct a total cost table for order quantities 100, 150, and 200 units.
(5 marks)

(d) Compute the expected cost for each order quantity and determine the optimal order quantity.
(3 marks)

(e) State two managerial insights from the obtained solution.
(1 mark)

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